

Question	Answer	Marks
1(a)	gravitational force (of attraction between satellite and planet)	B1
	<u>provides / is</u> centripetal force (on satellite about the planet)	B1
1(b)	$M = (4/3) \times \pi R^3 \rho$	B1
	$\omega = 2\pi / T$ or $v = 2\pi nR / T$	B1
	$GM / (nR)^2 = nR\omega^2$ or v^2 / nR	M1
	substitution clear to give $\rho = 3\pi n^3 / GT^2$	A1
1(c)	$n = (3.84 \times 10^5) / (6.38 \times 10^3) = 60.19$ or 60.2	C1
	$\rho = 3\pi \times 60.19^3 / [(6.67 \times 10^{-11}) \times (27.3 \times 24 \times 3600)^2]$	C1
	$\rho = 5.54 \times 10^3 \text{ kg m}^{-3}$	A1